

Cost Analysis

2026-05-11

Contents

1	Assumptions	1
2	Executive Summary	2
3	Breakdown by grades	10
3.1	6010085	10
3.2	3200115	14
3.3	6010100	18
3.4	6010120	22

1 Assumptions

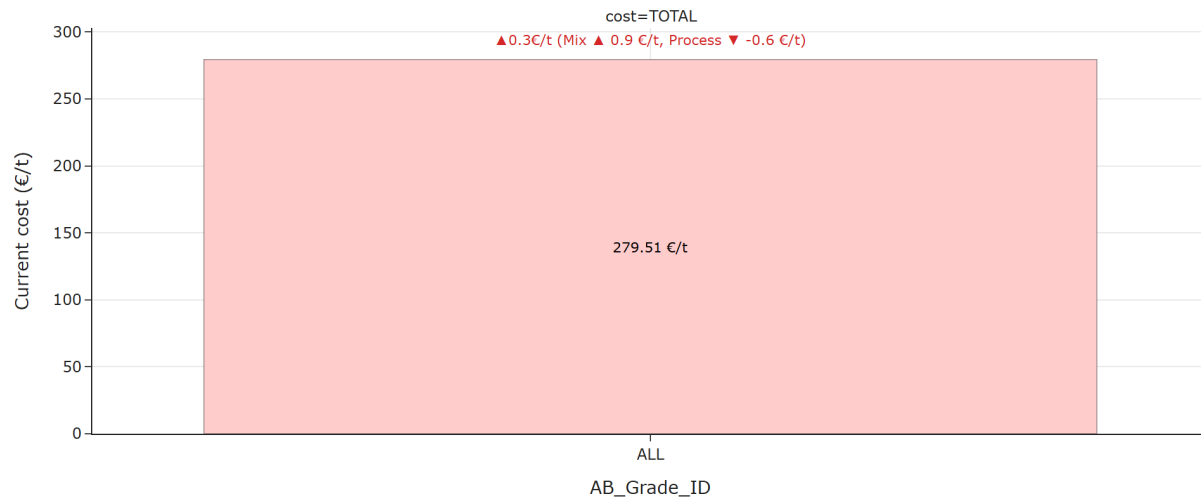
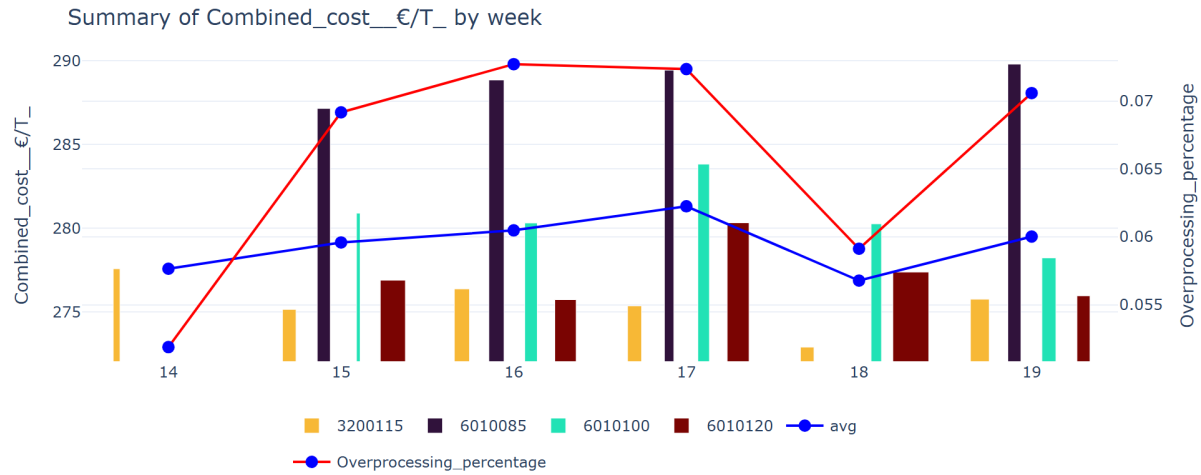
The aim of the analysis is to identify the most cost-effective production periods in months of data and recommend optimal operating windows for the corresponding key influencers that can be set as a benchmark for the weekly analysis.

- **Period Analyzed:** April 05, 2026 – May 11, 2026
- **Cost Components Included:** Fibre, steam, electricity, starch, and chemicals.
- **Grades Included:** Only grades 6010085, 6010100, 601020 and 3200115 are included in this week’s analysis.
- **Data Selection Criteria:** Only operating conditions where strength properties exceed specification thresholds were considered.
- **Naming Conventions Used:**
 - **Current:** Refers to the last week of the analyzed period.
 - **Historic:** Represents the average cost over the monthly interval previous to the “Current” period. Used as baseline.
 - **Best:** Corresponds to the cost associated with the optimal batch configuration.

2 Executive Summary

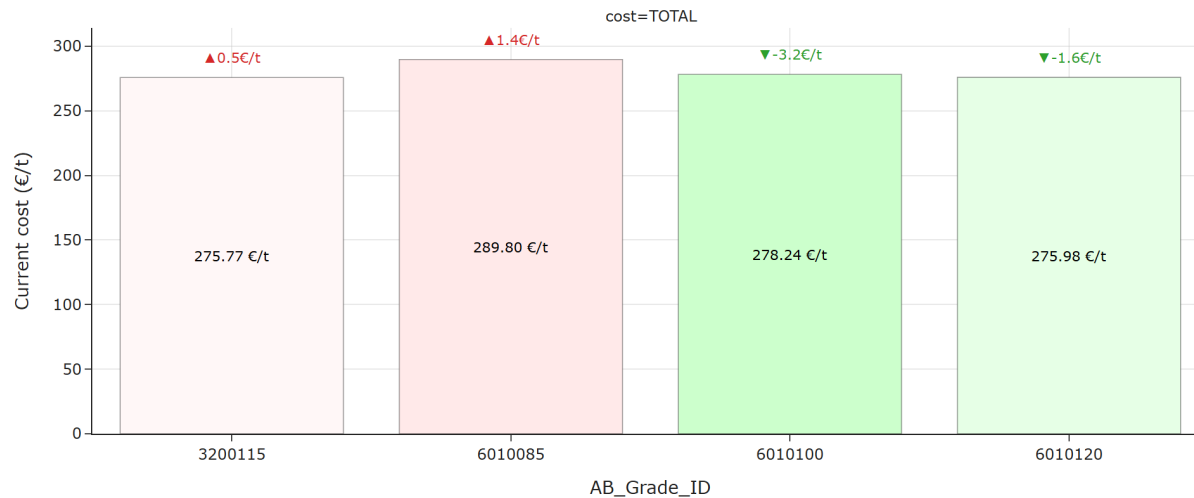
The following plot illustrates how total cost varies across different grades, enabling a comparison between the current configuration and the historical baseline.

The next plot displays how total cost evolves over time, with a breakdown by grade to show individual contributions.



For all considered grades, TOTAL cost saw an increase of 0.26 €/t, resulting in 279.51 €/t.

Breaking down the overall cost change, +0.91 €/t was driven by a more expensive mix of grades, and -0.65 €/t was due to an improvement in the process.



For grade 3200115, TOTAL cost saw an increase of 0.49 €/t, resulting in 275.77 €/t.

For grade 6010085, TOTAL cost saw an increase of 1.39 €/t, resulting in 289.80 €/t.

For grade 6010100, TOTAL cost saw a decrease of 3.19 €/t, resulting in 278.24 €/t.

For grade 6010120, TOTAL cost saw a decrease of 1.58 €/t, resulting in 275.98 €/t.

Looking across individual grades, the strongest cost increase was observed for grade 6010085 (+1.39 €/t) and the strongest cost decrease was observed for grade 6010100 (-3.19 €/t). These movements stand out and may require further investigation.



Average change by component:

- **Starch_€/T** decreased on average (-0.56 €/t).
- **Steam_€/T** decreased on average (-0.45 €/t).
- **Electricity_€/T** increased on average (+0.15 €/t).
- **Fibre_cost_€/T** increased on average (+0.13 €/t).
- **Chemicals_€/T** increased on average (+0.02 €/t).

A more detailed breakdown by component shows that the largest average increase comes from **Electricity_€/T** (+0.15 €/t on average), with the highest increase in grade 6010100 (+0.58 €/t) and the largest average decrease comes from **Starch_€/T** (-0.56 €/t on average), with the biggest drop in grade 6010100 (-1.85 €/t). These components stand out and may require further investigation.



Average change by steam:

- **Steam_Whitewater** decreased on average (-0.68 €/t).
- **Steam_Afterdryers** increased on average (+0.29 €/t).
- **Steam_Predryers** decreased on average (-0.07 €/t).
- **Steam_Starchkitchen** increased on average (+0.01 €/t).
- **Steam_Heatexchangers** decreased on average (-0.01 €/t).

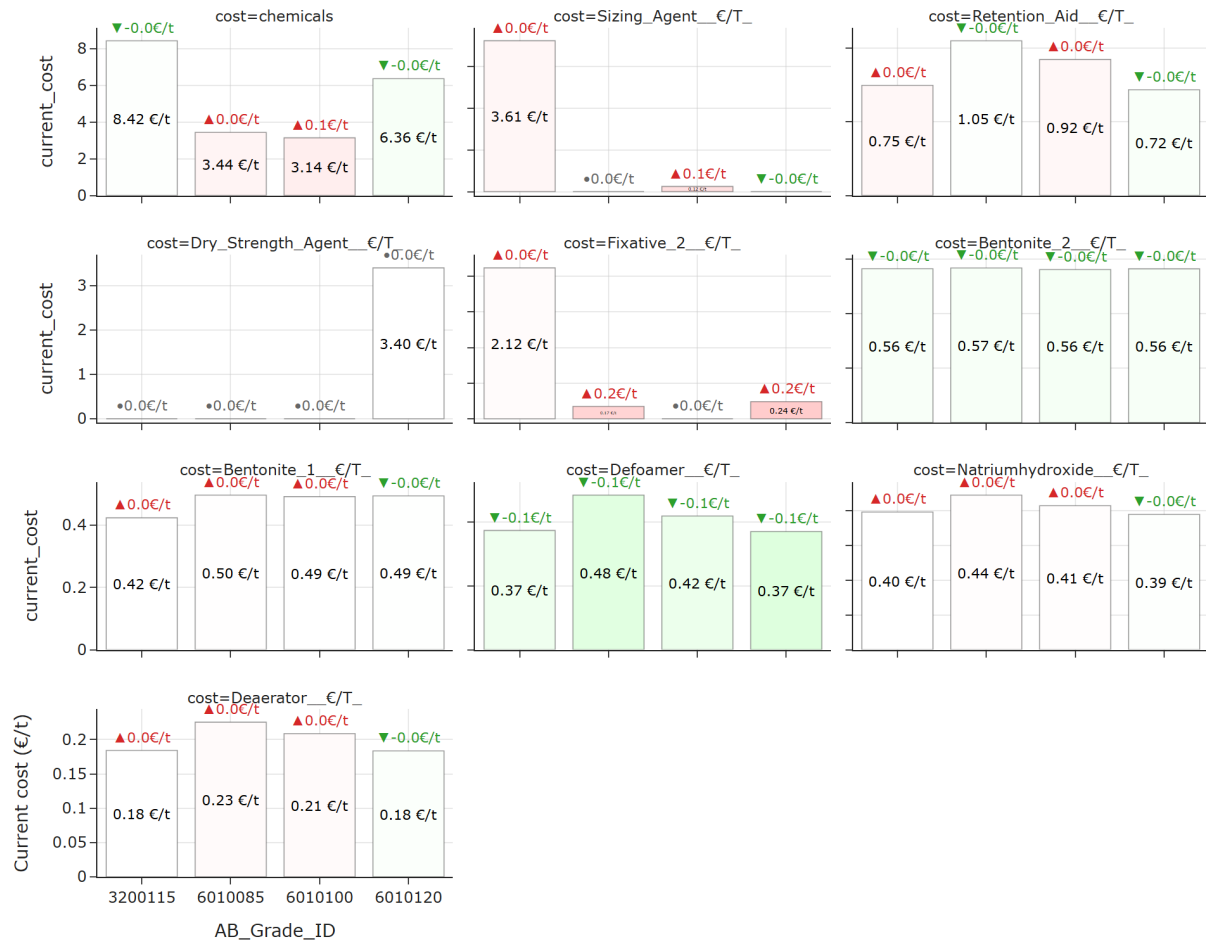
A more detailed breakdown by component shows that the largest average increase comes from **Steam_Afterdryers** (+0.29 €/t on average), with the highest increase in grade 3200115 (+1.05 €/t) and the largest average decrease comes from **Steam_Whitewater** (-0.68 €/t on average), with the biggest drop in grade 6010100 (-1.25 €/t). These components stand out and may require further investigation.



Average change by electricity:

- **Electricity_Specific** increased on average (+0.10 €/t).
- **Electricity_Other** increased on average (+0.06 €/t).
- **Electricity_Press_and_Steambox** increased on average (+0.05 €/t).
- **Electricity_Wiresection** decreased on average (-0.05 €/t).
- **Electricity_Afterdryer** decreased on average (-0.03 €/t).
- **Electricity_Freqconverters** increased on average (+0.01 €/t).
- **Electricity_Predryer** increased on average (+0.00 €/t).

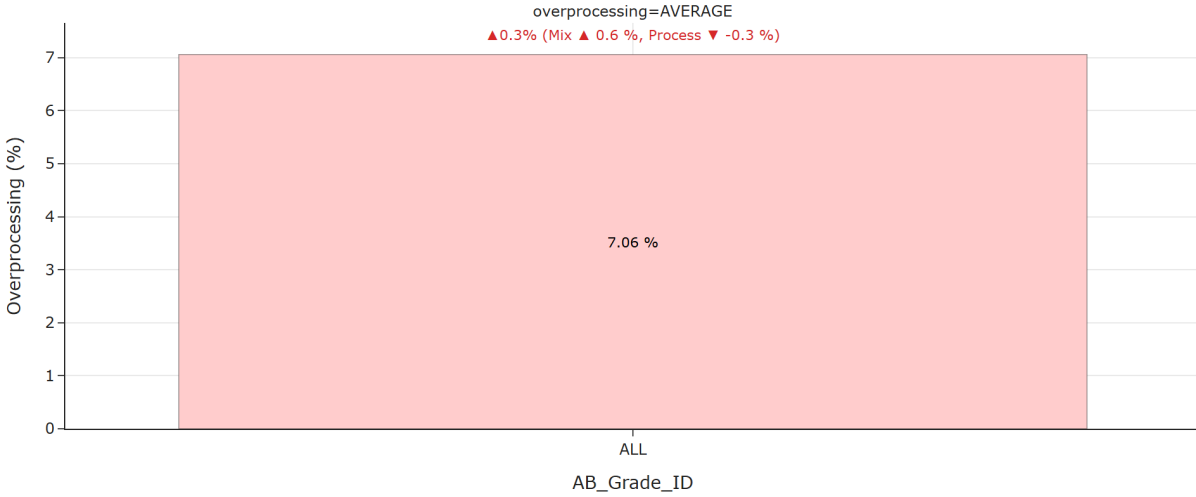
A more detailed breakdown by component shows that the largest average increase comes from **Electricity_Specific** (+0.10 €/t on average), with the highest increase in grade 6010100 (+0.38 €/t) and the largest average decrease comes from **Electricity_Wiresection** (-0.05 €/t on average), with the biggest drop in grade 3200115 (-0.07 €/t). These components stand out and may require further investigation.



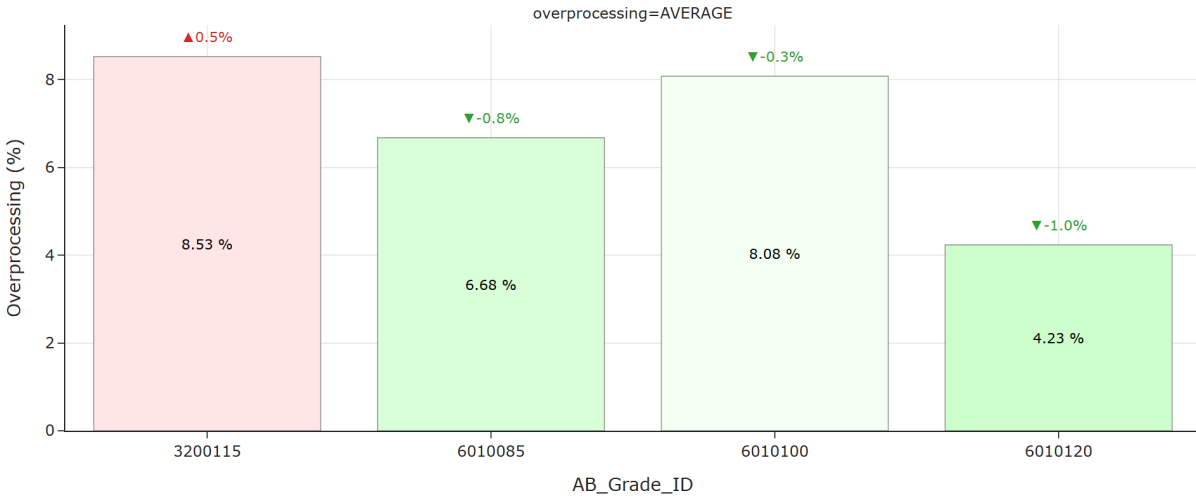
Average change by chemicals:

- **Fixative_2_€/T** increased on average (+0.13 €/t).
- **Defoamer_€/T** decreased on average (-0.10 €/t).
- **Sizing_Agent_€/T** increased on average (+0.08 €/t).
- **Bentonite_2_€/T** decreased on average (-0.03 €/t).
- **Retention_Aid_€/T** increased on average (+0.01 €/t).
- **Deaerator_€/T** increased on average (+0.01 €/t).
- **Natriumhydroxide_€/T** increased on average (+0.00 €/t).
- **Bentonite_1_€/T** increased on average (+0.00 €/t).
- **Dry_Strength_Agent_€/T** was unchanged on average (0.00 €/t).

A more detailed breakdown by component shows that the largest average increase comes from **Fixative_2_€/*T*** (+0.13 €/t on average), with the highest increase in grade 6010120 (+0.21 €/t) and the largest average decrease comes from **Defoamer_€/*T*** (-0.10 €/t on average), with the biggest drop in grade 6010120 (-0.13 €/t). These components stand out and may require further investigation.



For all considered grades, TOTAL overprocessing saw an increase of 0.27 %, resulting in 7.06 %.



For grade 3200115, TOTAL overprocessing saw an increase of 0.50 %, resulting in 8.53 %.

For grade 6010085, TOTAL overprocessing saw a decrease of 0.78 %, resulting in 6.68 %.

For grade 6010100, TOTAL overprocessing saw a decrease of 0.25 %, resulting in 8.08 %.

For grade 6010120, TOTAL overprocessing saw a decrease of 1.01 %, resulting in 4.23 %.

Looking across individual grades, the strongest overprocessing increase was observed for grade

3200115 (+0.50 %) and the strongest overprocessing decrease was observed for grade 6010120 (-1.01 %). These movements stand out and may require further investigation.

AttributeError: 'DataFrame' object has no attribute 'cost'. Did you mean: 'count'?



For grade 3200115, TOTAL overprocessing saw an increase of 0.50 %, resulting in 8.53 %.

For grade 6010085, TOTAL overprocessing saw a decrease of 0.78 %, resulting in 6.68 %.

For grade 6010100, TOTAL overprocessing saw a decrease of 0.25 %, resulting in 8.08 %.

For grade 6010120, TOTAL overprocessing saw a decrease of 1.01 %, resulting in 4.23 %.

Looking across individual grades, the strongest overprocessing increase was observed for grade 3200115 (+0.50 %) and the strongest overprocessing decrease was observed for grade 6010120 (-1.01 %). These movements stand out and may require further investigation.

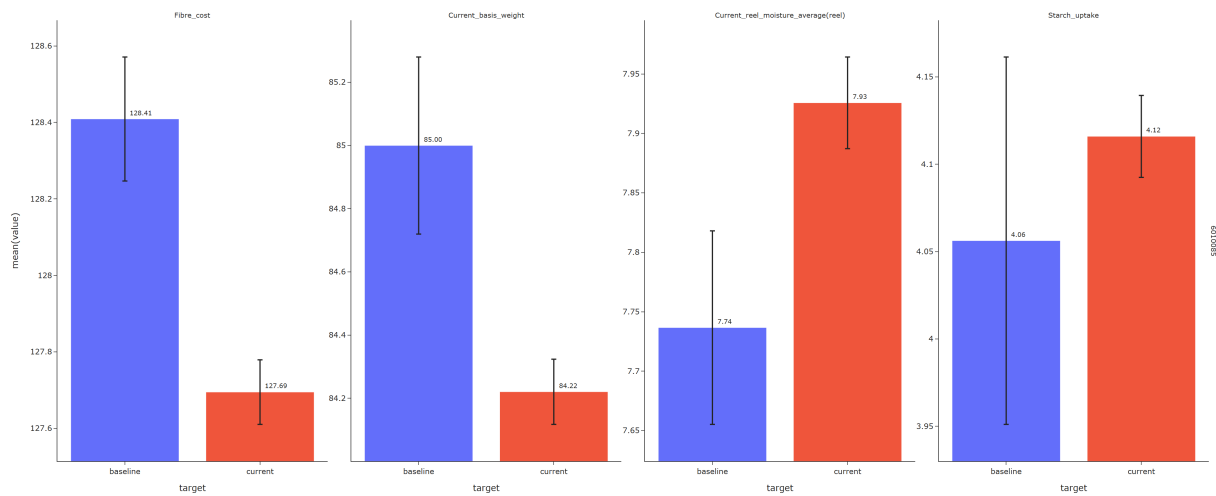
3 Breakdown by grades

3.1 6010085

3.1.1 Fiber cost



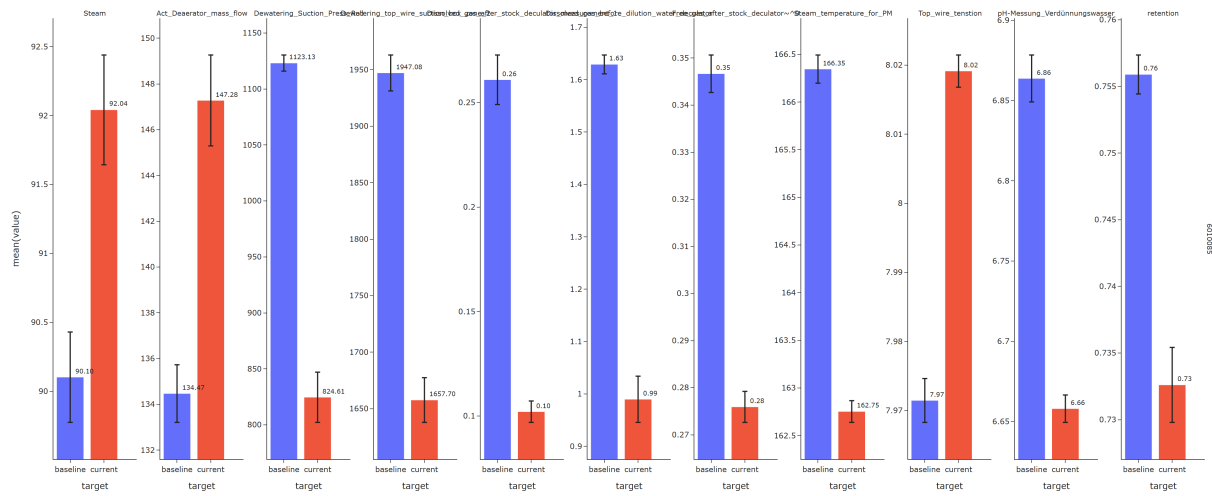
For grade 6010085, the cost change is mainly explained by the following factors (top 20% by contribution): Cost-reducing drivers: - Current reel moisture average(reel) increased (+0.19)



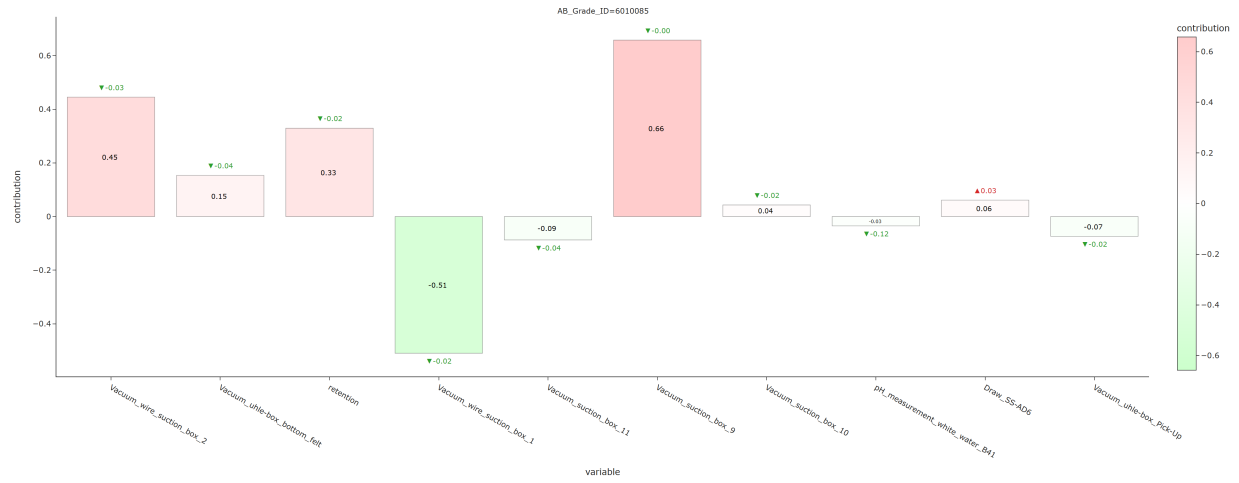
3.1.2 Steam cost



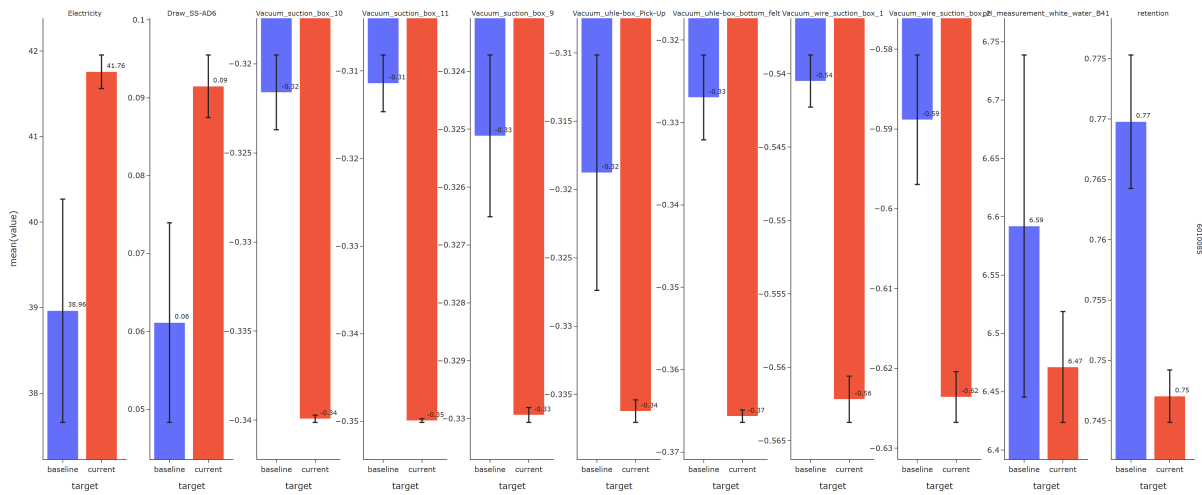
For grade 6010085, the cost change is mainly explained by the following factors (top 20% by contribution): Cost-reducing drivers: - Free gas after stock decuator~0 decreased (-0.07) - Act Deaeator mass flow g/T increased (+12.81) - Top wire tension increased (+0.05) Cost-increasing drivers: - Dissolved gas before dilution water decuator decreased (-0.64) - Dissolved gas after stock decuator measurement 1 decreased (-0.16)



3.1.3 Electricity cost



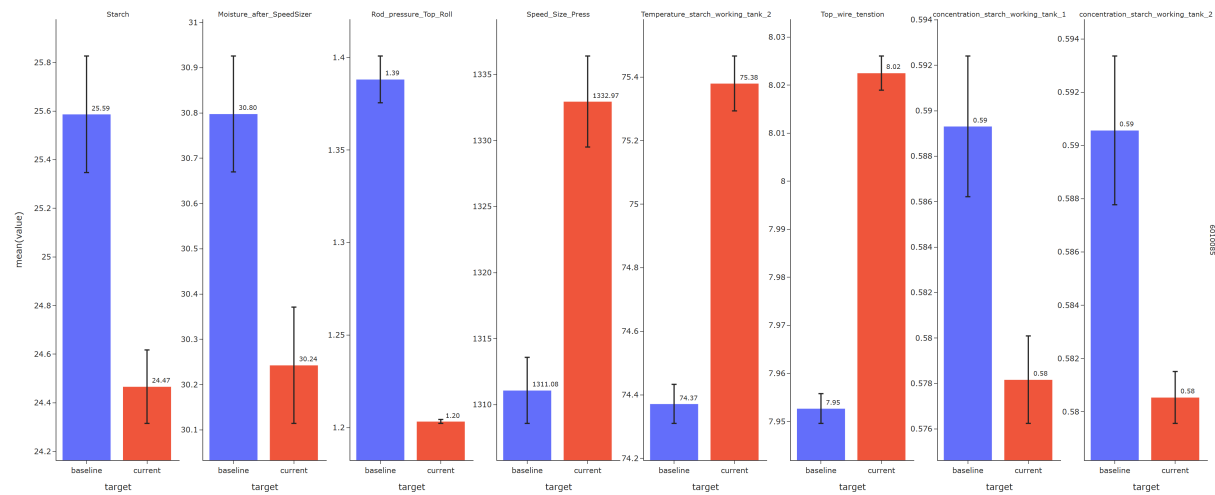
For grade 6010085, the cost change is mainly explained by the following factors (top 20% by contribution): Cost-reducing drivers: - Vacuum wire suction box 1 decreased (-0.02) Cost-increasing drivers: - Vacuum suction box 9 decreased (-0.00) - Vacuum wire suction box 2 decreased (-0.03)



3.1.4 Starch cost



For grade 6010085, the cost change is mainly explained by the following factors (top 20% by contribution): Cost-reducing drivers: - Moisture after SpeedSizer decreased (-0.56) - Concentration starch working tank 1 decreased (-0.01)

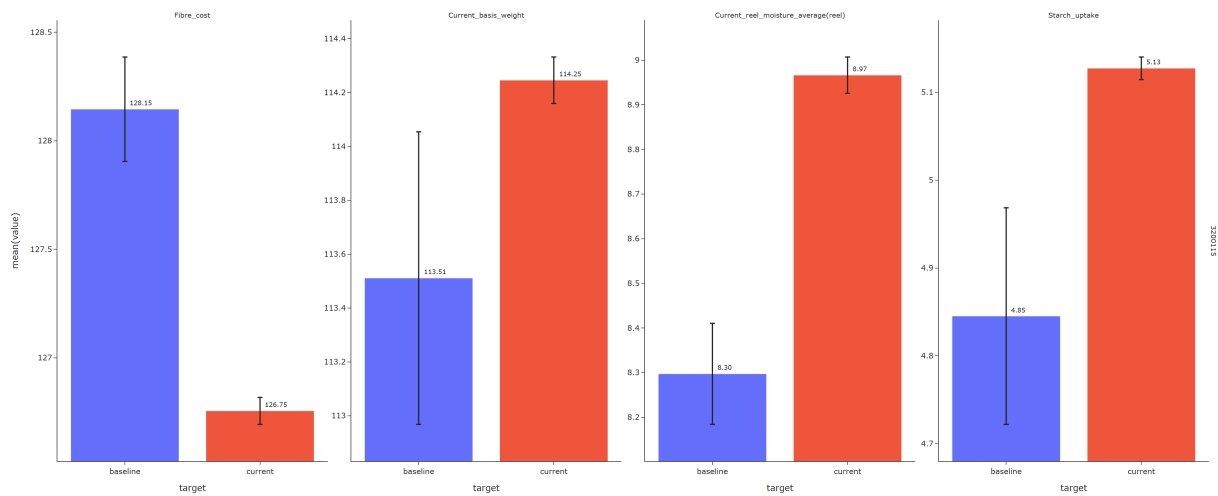


3.2 3200115

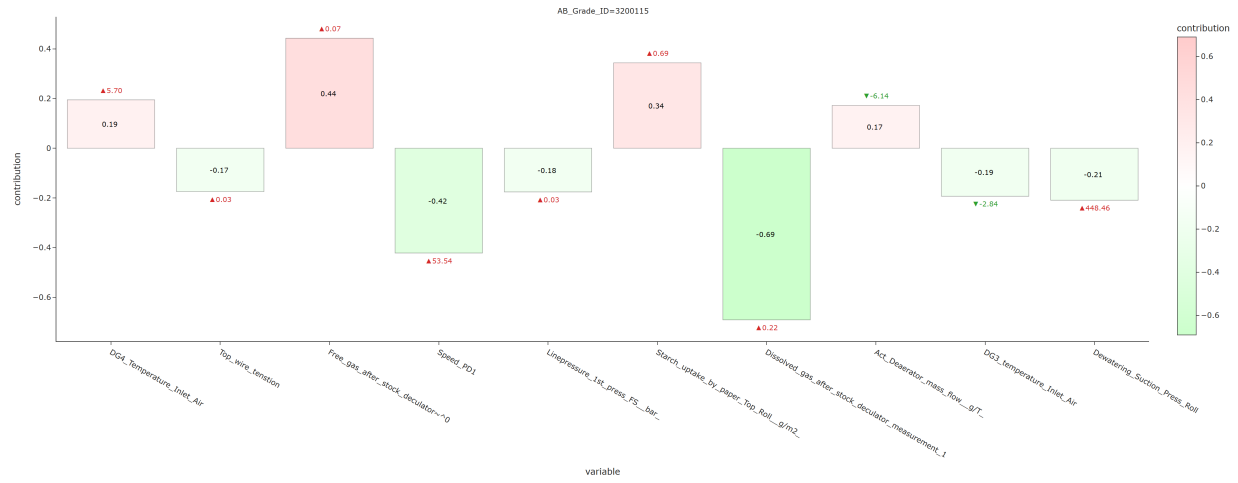
3.2.1 Fiber cost



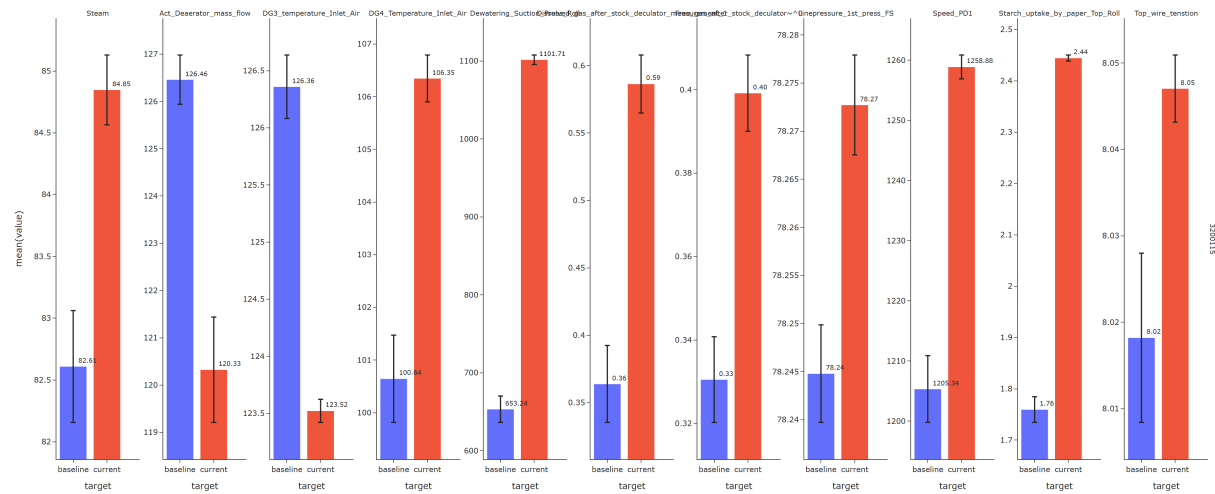
For grade 3200115, the cost change is mainly explained by the following factors (top 20% by contribution): Cost-reducing drivers: - Current reel moisture average(reel) increased (+0.67)



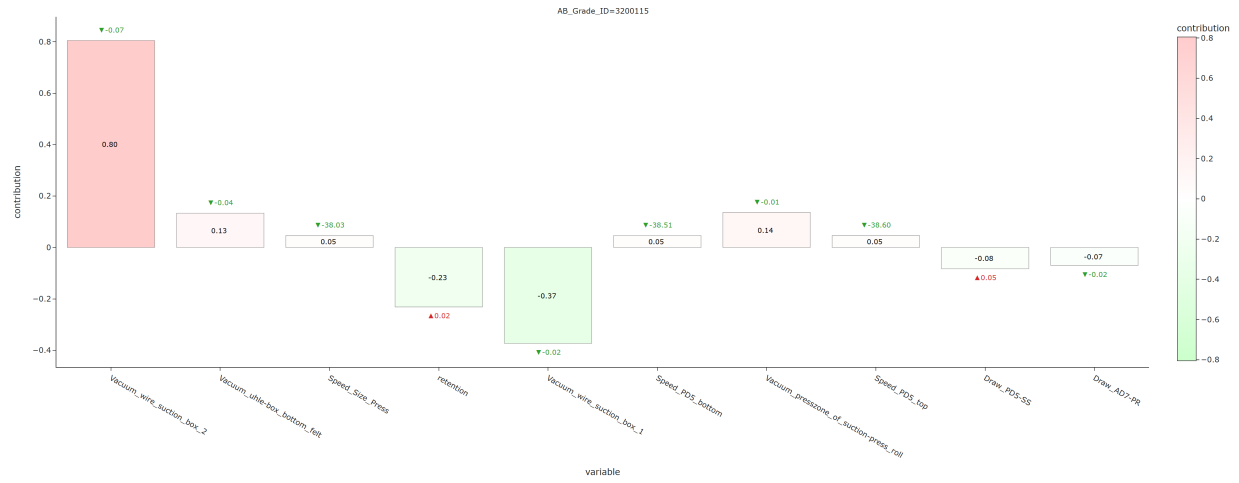
3.2.2 Steam cost



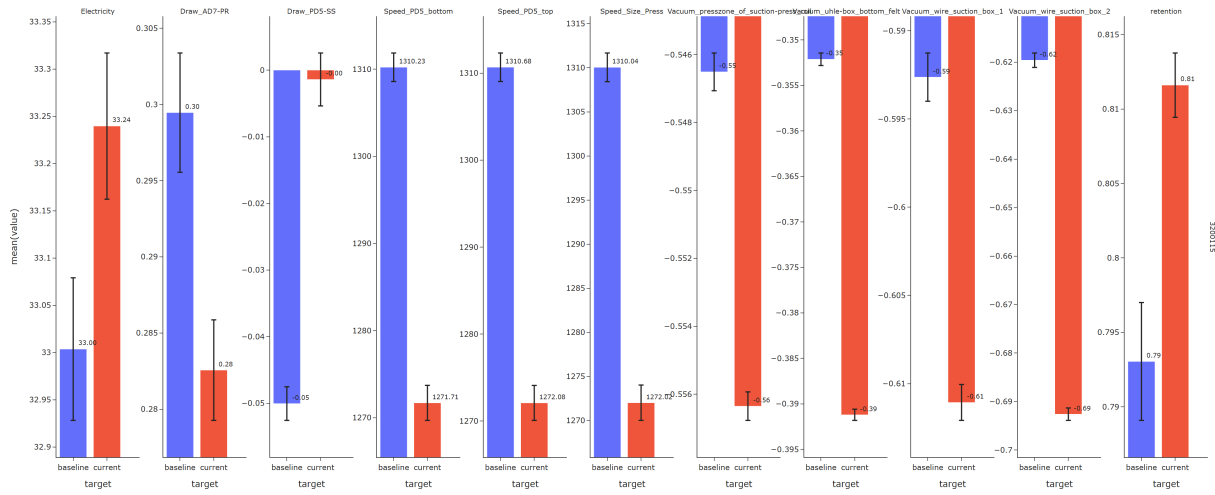
For grade 3200115, the cost change is mainly explained by the following factors (top 20% by contribution): Cost-reducing drivers: - Dissolved gas after stock deculator measurement 1 increased (+0.22) - Speed PD1 increased (+53.54) - Dewatering Suction Press Roll increased (+448.46) Cost-increasing drivers: - Free gas after stock decuator~0 increased (+0.07) - Starch uptake by paper Top Roll g/m2 increased (+0.69) - DG4 Temperature Inlet Air increased (+5.70)



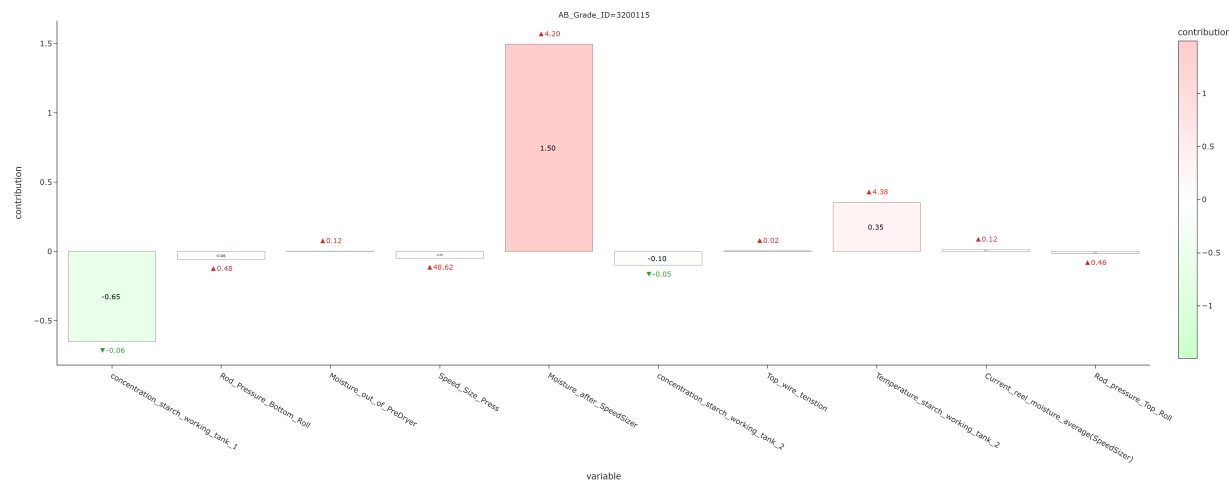
3.2.3 Electricity cost



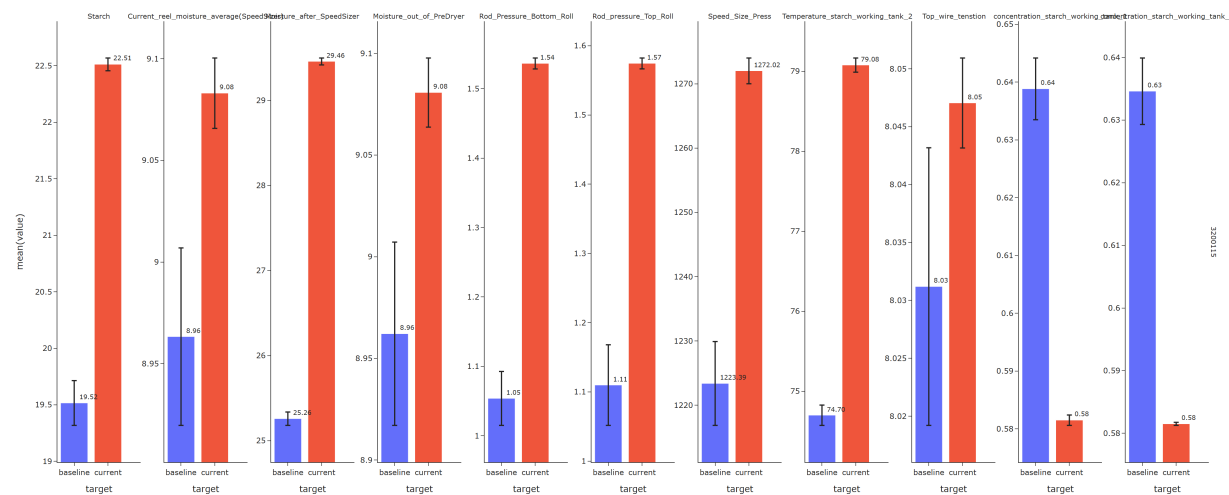
For grade 3200115, the cost change is mainly explained by the following factors (top 20% by contribution): Cost-reducing drivers: - Vacuum wire suction box 1 decreased (-0.02) - Retention increased (+0.02) Cost-increasing drivers: - Vacuum wire suction box 2 decreased (-0.07) - Vacuum presszone of suction-press roll decreased (-0.01) - Vacuum uhle-box bottom felt decreased (-0.04)



3.2.4 Starch cost

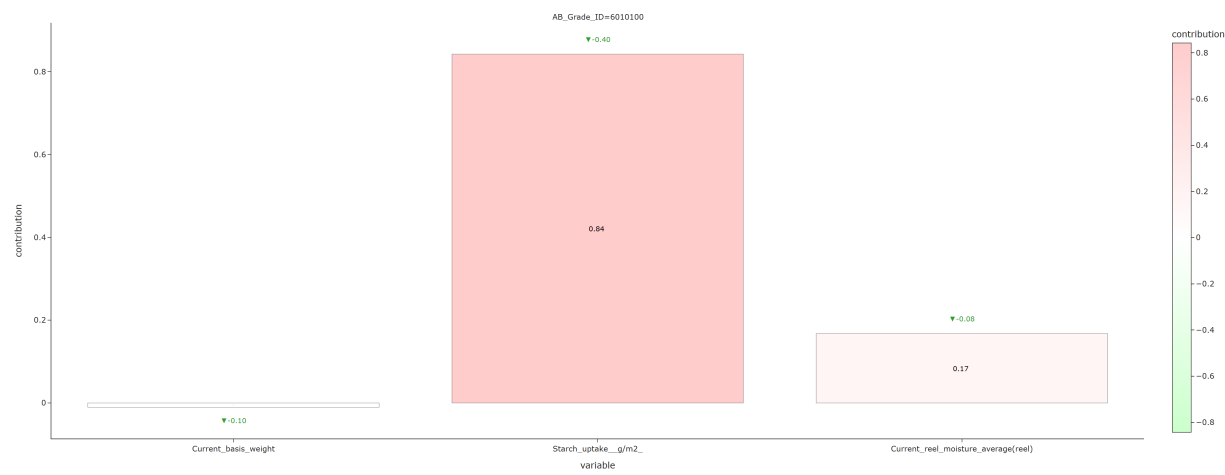


For grade 3200115, the cost change is mainly explained by the following factors (top 20% by contribution): Cost-reducing drivers: - Concentration starch working tank 1 decreased (-0.06)
Cost-increasing drivers: - Moisture after SpeedSizer increased (+4.20)

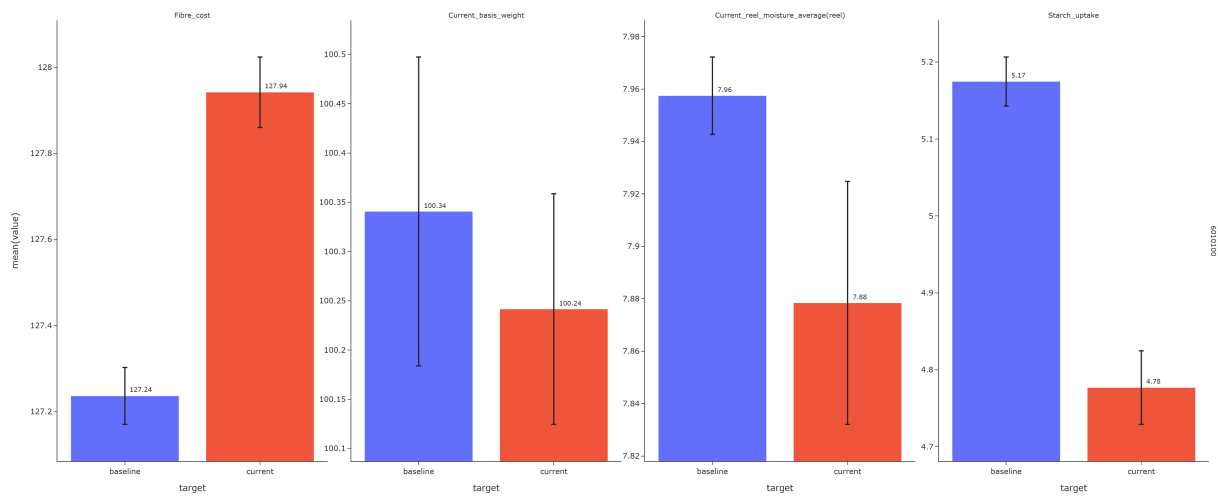


3.3 6010100

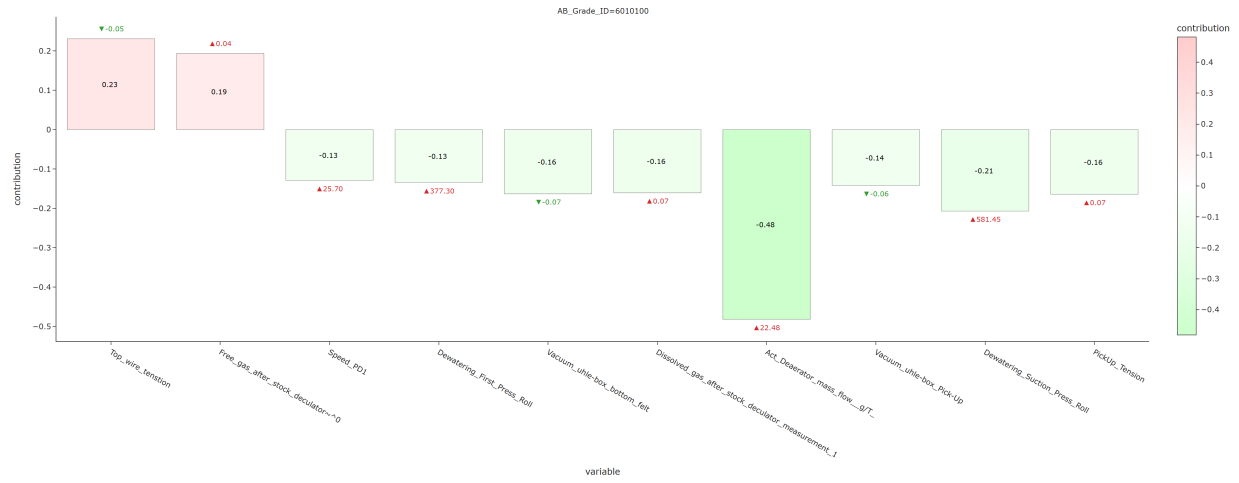
3.3.1 Fiber cost



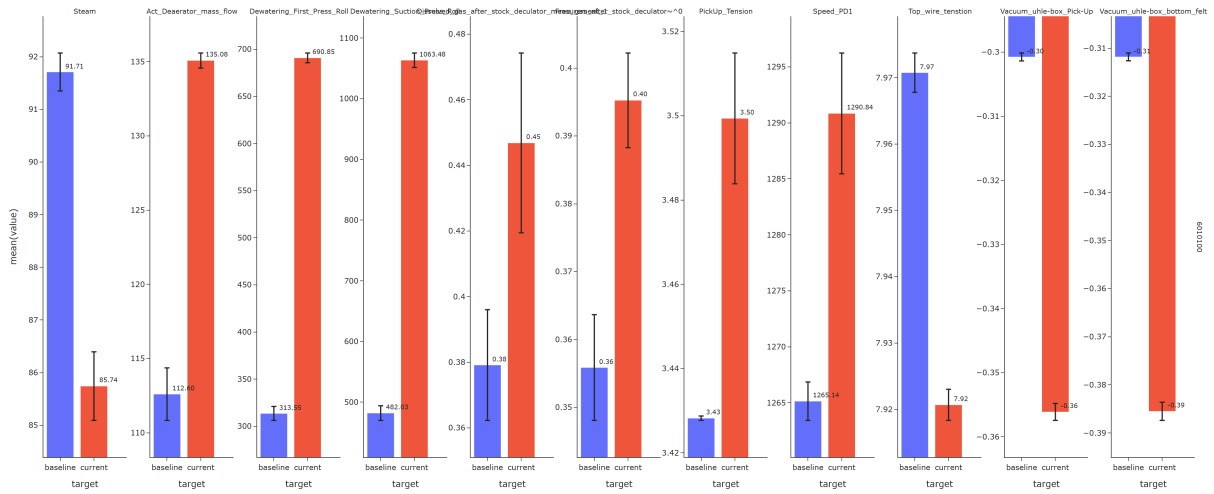
For grade 6010100, the cost change is mainly explained by the following factors (top 20% by contribution): Cost-increasing drivers: - Starch uptake g/m2 decreased (-0.40)



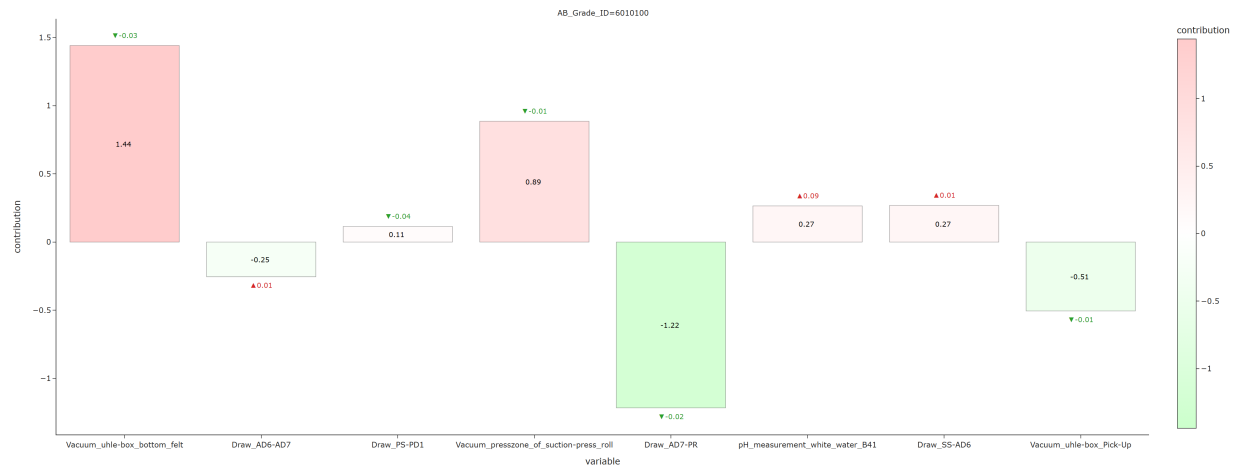
3.3.2 Steam cost



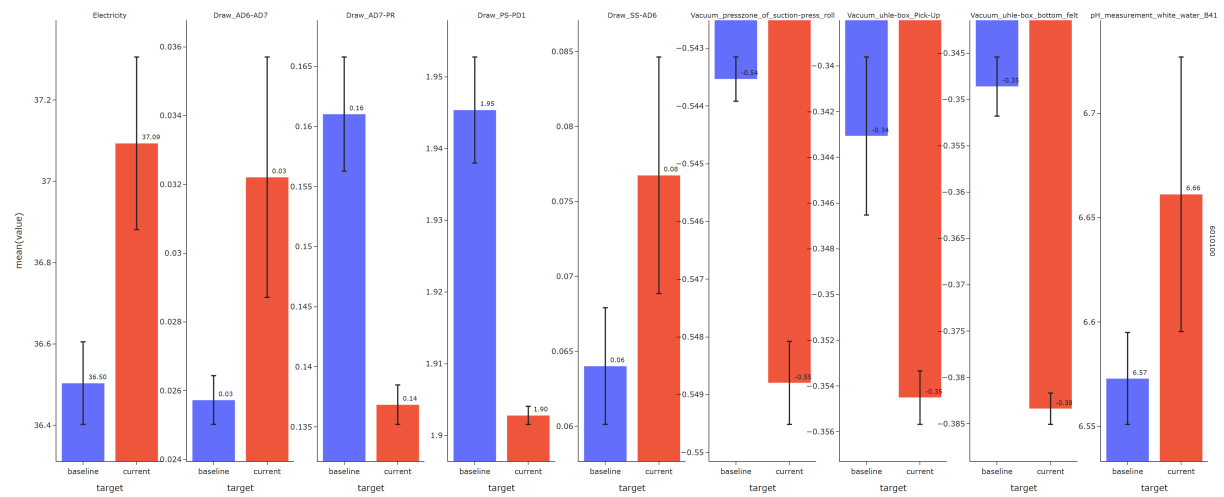
For grade 6010100, the cost change is mainly explained by the following factors (top 20% by contribution): Cost-reducing drivers: - Act Deaerator mass flow g/T increased (+22.48) - Dewatering Suction Press Roll increased (+581.45) - PickUp Tension increased (+0.07) - Vacuum uhle-box bottom felt decreased (-0.07) Cost-increasing drivers: - Top wire tension decreased (-0.05) - Free gas after stock decuator~0 increased (+0.04)



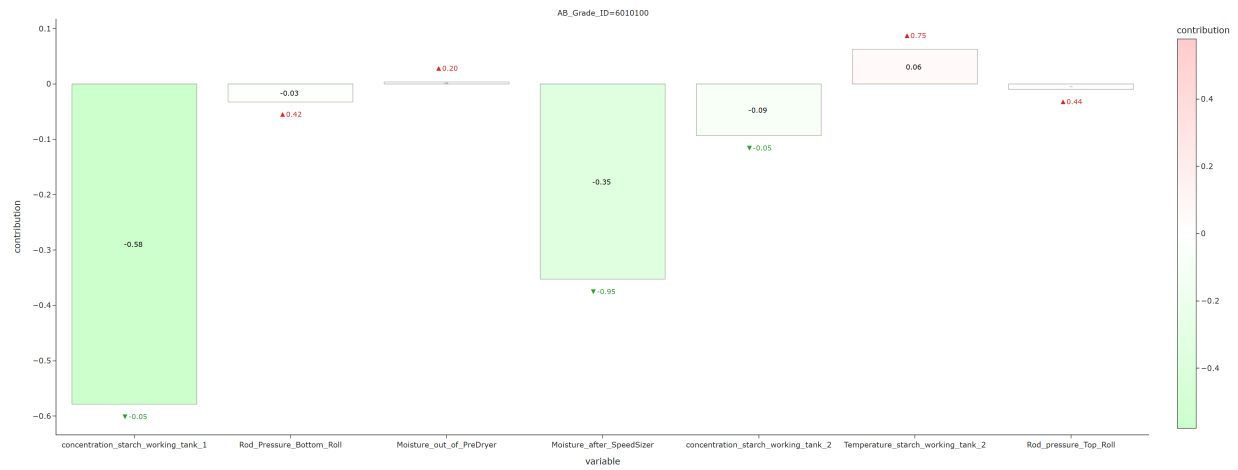
3.3.3 Electricity cost



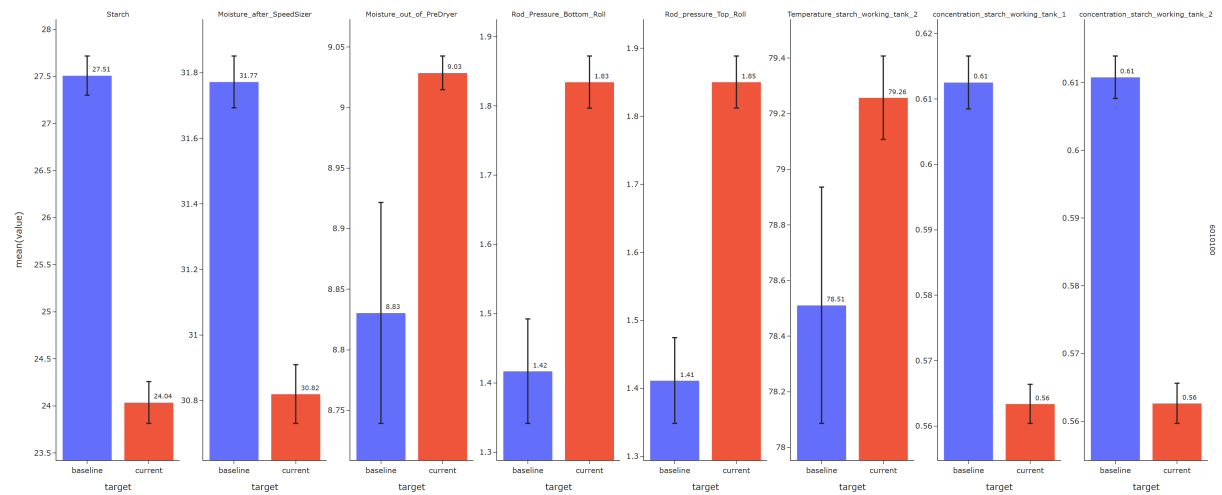
For grade 6010100, the cost change is mainly explained by the following factors (top 20% by contribution): Cost-reducing drivers: - Draw AD7-PR decreased (-0.02) Cost-increasing drivers: - Vacuum uhle-box bottom felt decreased (-0.03)



3.3.4 Starch cost

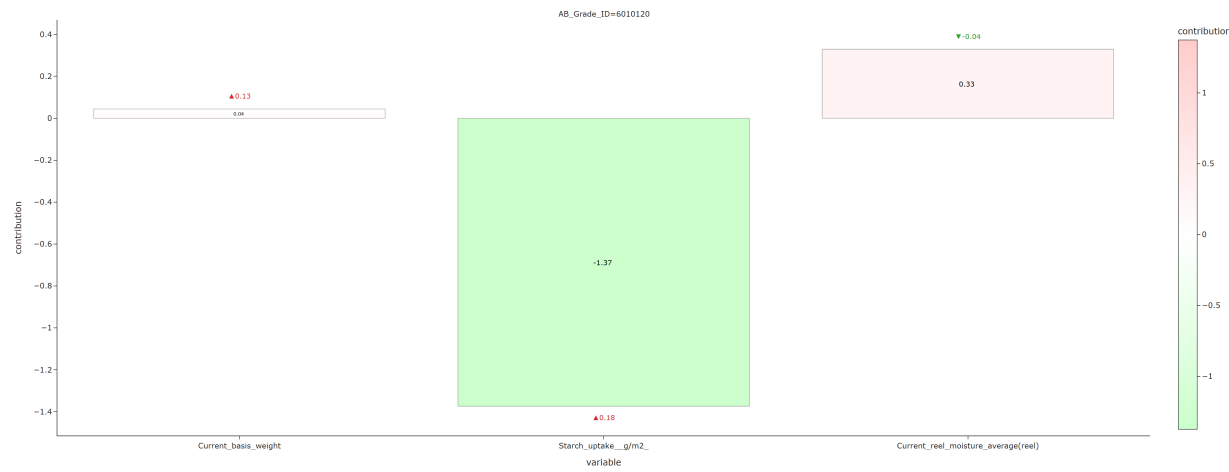


For grade 6010100, the cost change is mainly explained by the following factors (top 20% by contribution): Cost-reducing drivers: - Concentration starch working tank 1 decreased (-0.05) - Moisture after SpeedSizer decreased (-0.95)

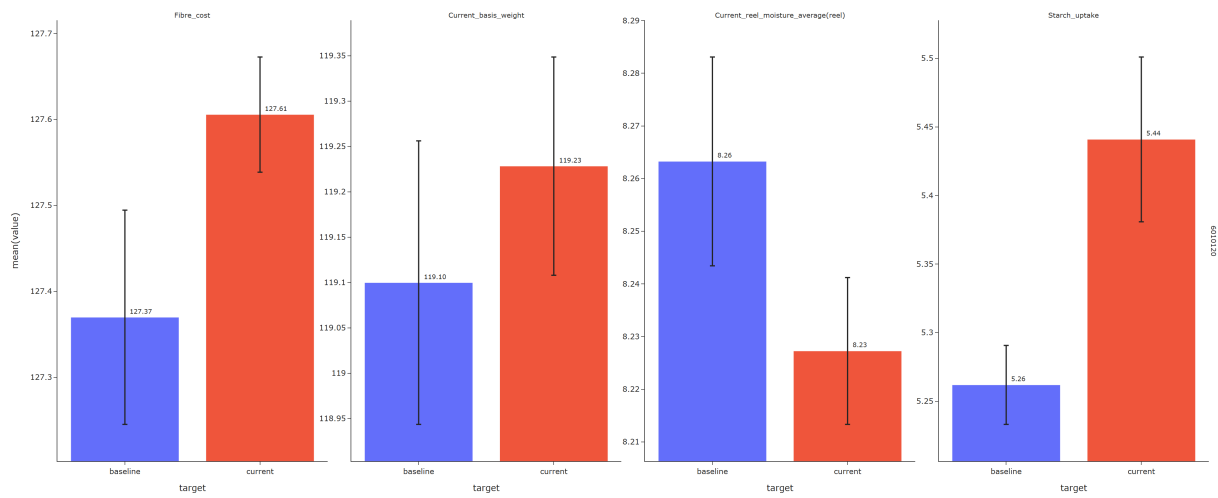


3.4 6010120

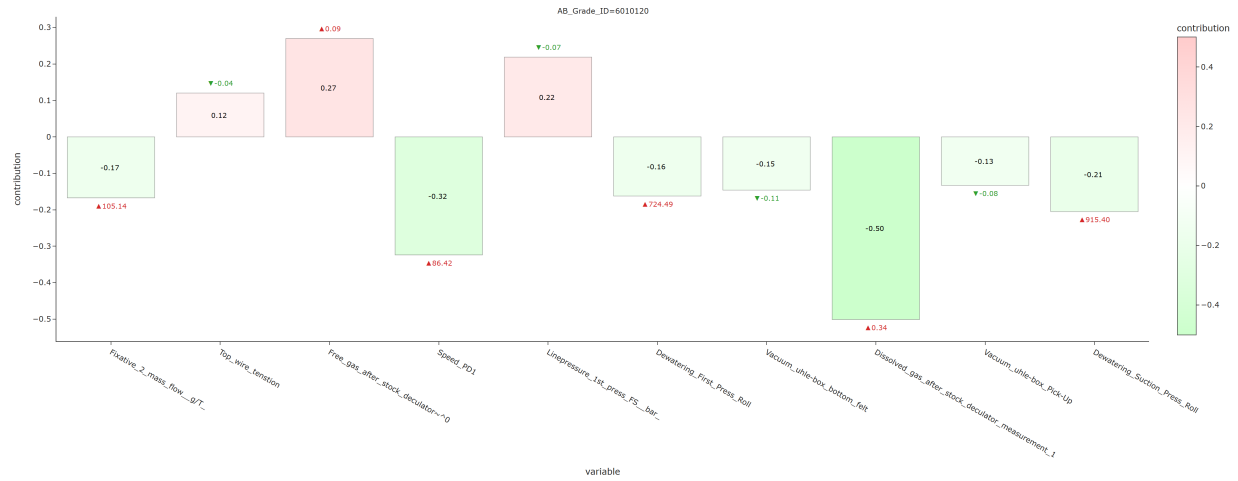
3.4.1 Fiber cost



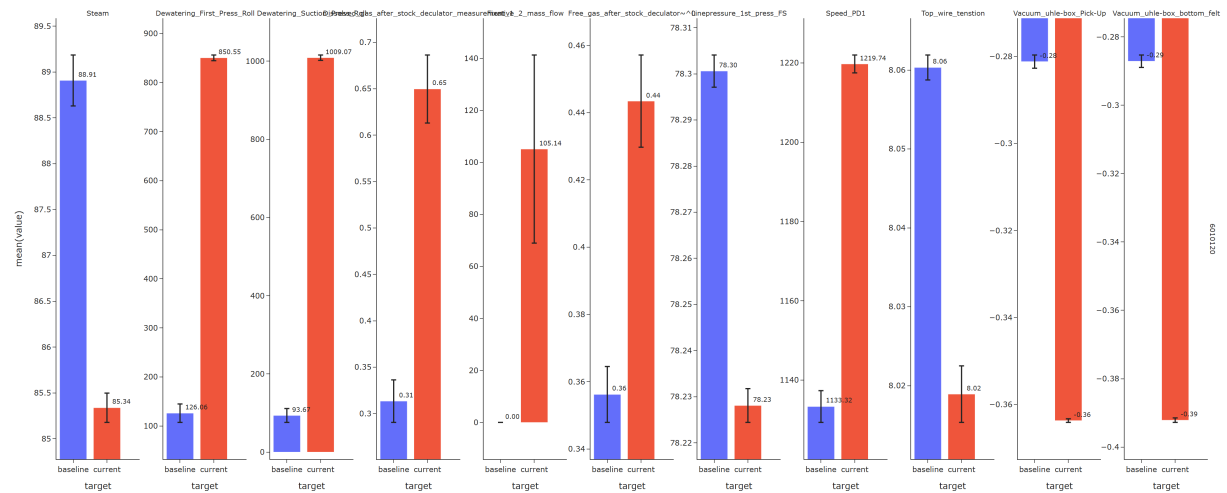
For grade 6010120, the cost change is mainly explained by the following factors (top 20% by contribution): Cost-reducing drivers: - Starch uptake g/m2 increased (+0.18)



3.4.2 Steam cost



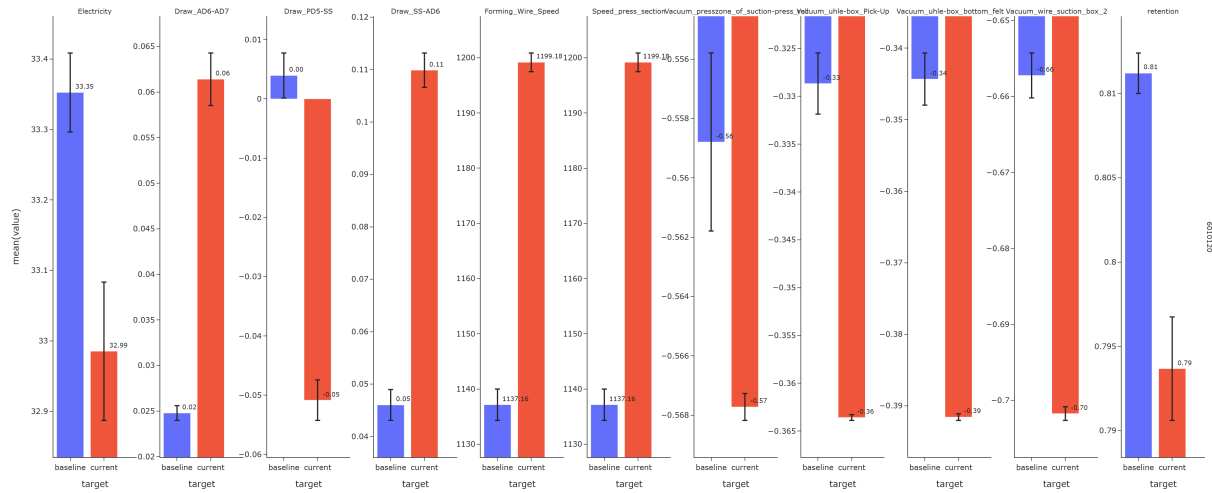
For grade 6010120, the cost change is mainly explained by the following factors (top 20% by contribution): Cost-reducing drivers: - Dissolved gas after stock deculator measurement 1 increased (+0.34) - Speed PD1 increased (+86.42) - Dewatering Suction Press Roll increased (+915.40) - Fixative 2 mass flow g/T increased (+105.14) - Dewatering First Press Roll increased (+724.49) Cost-increasing drivers: - Free gas after stock decuator~0 increased (+0.09) - Linepressure 1st press FS bar decreased (-0.07)



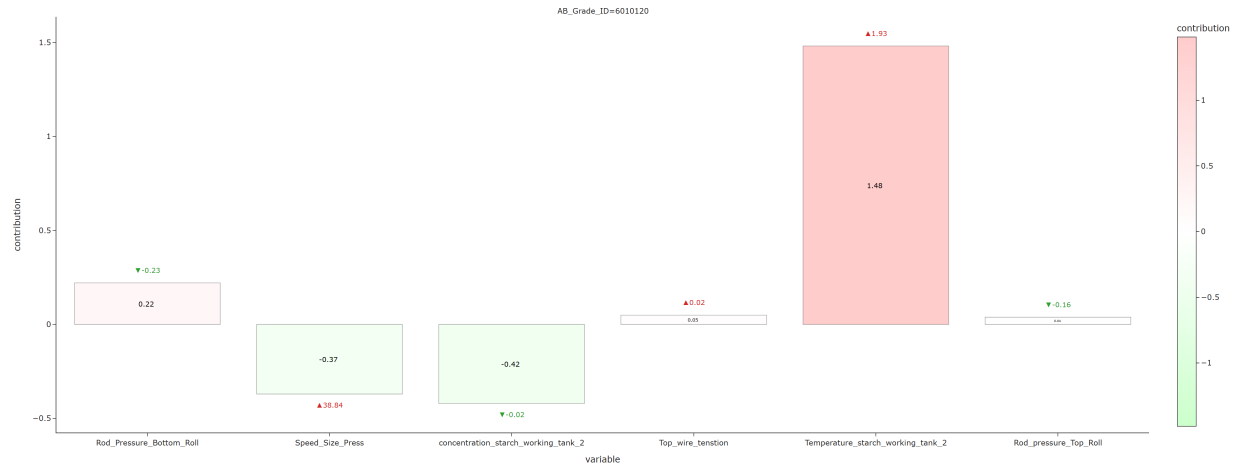
3.4.3 Electricity cost



For grade 6010120, the cost change is mainly explained by the following factors (top 20% by contribution): Cost-reducing drivers: - Vacuum uhle-box Pick-Up decreased (-0.03) Cost-increasing drivers: - Vacuum wire suction box 2 decreased (-0.04) - Retention decreased (-0.02) - Vacuum uhle-box bottom felt decreased (-0.05) - Vacuum presszone of suction-press roll decreased (-0.01)



3.4.4 Starch cost



For grade 6010120, the cost change is mainly explained by the following factors (top 20% by contribution): Cost-reducing drivers: - Concentration starch working tank 2 decreased (-0.02)
Cost-increasing drivers: - Temperature starch working tank 2 increased (+1.93)

